

Pagina 1: Visao Geral do Sistema

Publico

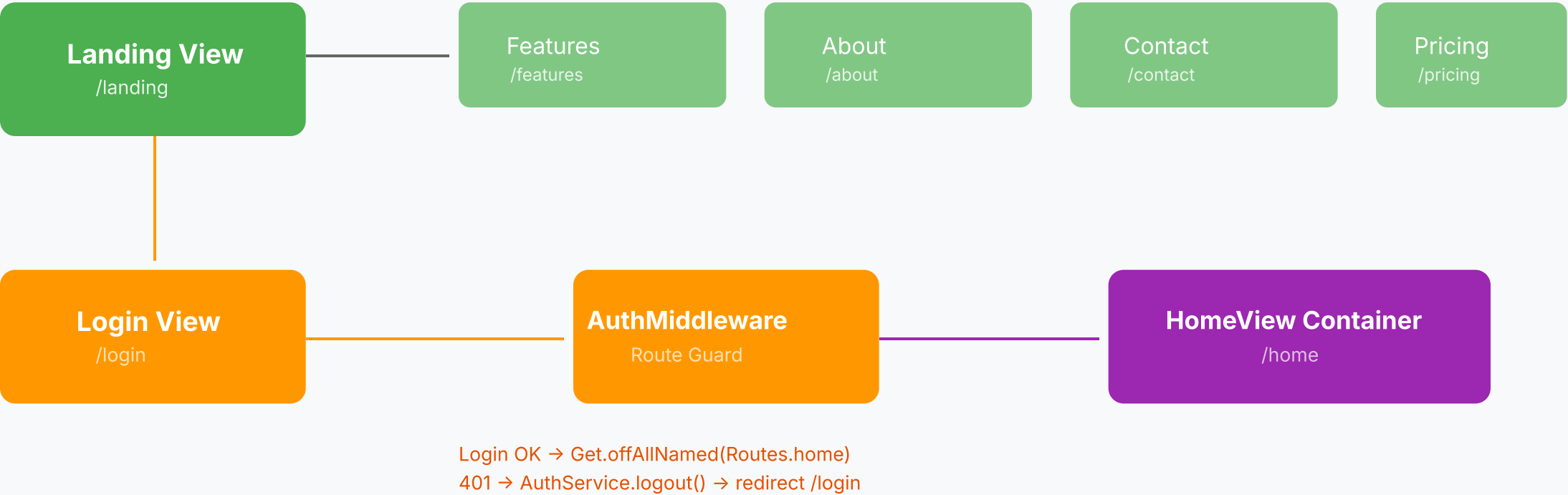
Autenticacao

Protegido

Container

Fluxo

Fluxo de navegacao principal:



Total: 5 paginas publicas + Login + 17 modulos protegidos
Framework: Flutter + GetX + Dio + Django REST API

App inicializa em /landing (GetMaterialApp initialRoute)

Rotas protegidas usam AuthMiddleware que verifica AuthService.isAuthenticated

Token armazenado em FlutterSecureStorage, anexado via Dio interceptor

main.dart init order:

1. GetStorage.init() 2. ApiService() 3. AuthService().init()

4. SettingsController() 5. GetMaterialApp(theme, routes)

Pagina 3: Autenticacao e Seguranca

Arquitetura de autenticacao:



Fluxo detalhado:

- 1. Toda rota protegida dispara `AuthMiddleware.redirect()`
 - 2. Verifica `Get.find<AuthService>().isAuthenticated.value`
 - 3. Se FALSE → redirect para `/login`
 - 4. Se TRUE → retorna null (prossigue)
- Session expiry (ApiService Dio interceptor):**
OnError 401 → `AuthService.logout()` → redirect `/login` + snackbar
Token anexado em OnRequest via Authorization header
Dio config: timeout 60s, Content-Type application/json

Fluxo de Logout:

- 1. `HomeController.logout()` chama `AuthService.logout()`
- 2. `POST /api/logout/` (best-effort)
- 3. Deleta token do `FlutterSecureStorage`
- 4. `isAuthenticated = false` → `Get.offAllNamed(/login)`

Pagina 2: Area Publica e Login

Rotas publicas (sem middleware):

LandingView
/landing (initial)

FeaturesView
/features

AboutView
/about

ContactView
/contact

PricingView
/pricing

Rota de autenticacao:

LoginView
/login (public)

Fluxo de Login:

1. Usuario preenche email + senha

2. LoginController.login() chama AuthService.login()

3. POST /api/login/ com username/password

4. AuthService armazena token em FlutterSecureStorage

5. isAuthenticated = true → Get.offAllNamed(/home)

Landing page sections:

Hero Section - video background + CTAs

Features Section - 6 cards: Algoritmo, Alertas, Controle, Eficiencia, Visao, Historico

CTA Section - banner gradiente + botao Acessar Painel

Footer - WebFooter widget

Pagina 4: Area Protegida (Sidebar Home)

HomeView - Container Master com Sidebar:

HomeView

/home

Sidebar possui 4 secoes e 17 indices (0-16):

PRINCIPAL

Projeto (SiloViewer) [0]

Resumo IA (Dashboard) [1]

Smart Sense IA [9]

Processos [14]

MONITORAMENTO

U. Armazenadora [2]

Silos [3]

Lotes [12]

FINANCEIRO

Custos [16]

SISTEMA

Dispositivos [4]

Notificacoes [5]

Rotas protegidas com AuthMiddleware:

/home (HomeView - container master)
/support (SupportView)
/smart-sense-ia (SmartSenselAPIView)
/batch-management (BatchManagementView)
/secagem, /processos, /clientes, /custos
/devices/motors (fallback standalone)

Controle de Acesso:

Index 7 (AccessManagement) escondido para
nao-admins via controller.canManageUsers

HomeBinding injeta dependencias:

- HomeController
- U.ArmazenadoraController
- BatchManagementController
- SiloManagementController
- SecagemController
- ProcessosController
- ClientesController
- CustosController
- DashboardController

Pagina 5: Arquitetura de Dados e API

Camada de servicos e comunicacao com API:

ApiService
Dio HTTP Client

Configuracao Dio
Timeout 60s | JSON headers

Base URLs:
localhost:8000/api/
ai.secagemdigital.com/api/

Interceptors (Dio):

OnRequest: anexa Authorization: Token <token> header

OnError 401: AuthService.logout() + redirect /login

Key Methods (ApiService):

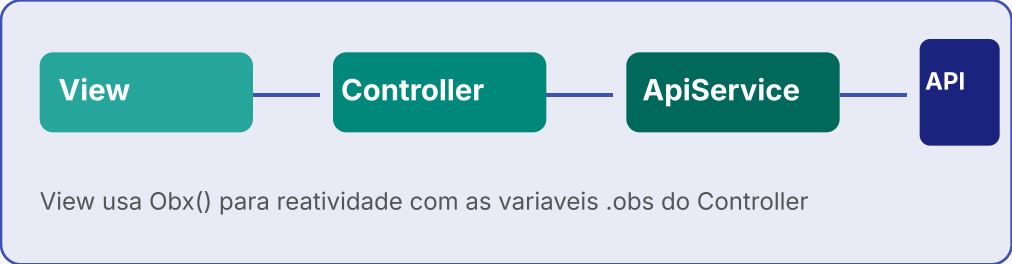
fetchContext() → GET /api/contexto/

postStream(endpoint, data) → SSE stream

extractRespostaOuResumo() → parse JSON

dio.get/post/put/delete → CRUD operations

Fluxo de Dados:



Servicos Core (GetXService):

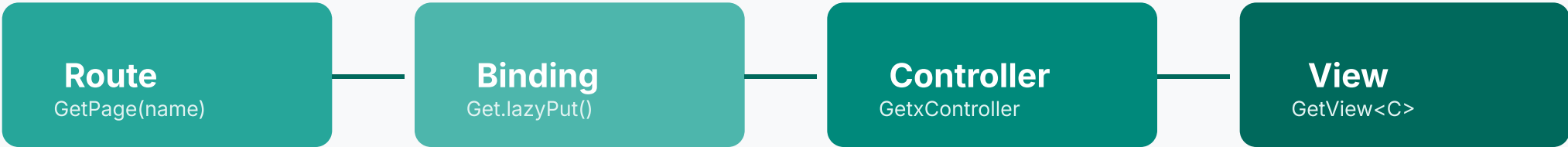
ApiService - Dio singleton, injetado em main.dart

AuthService - Token + isAuthenticated state

SettingsController - Theme + primary color

Pagina 6: GetX MVC + Notas da Arquitetura

Padrao GetX MVC (Arquitetura do Projeto):



Bindings injetam Controllers; Controllers expõem Rx (.obs); Views usam Obx()

Notas da Arquitetura:

- 1. Feature-first modular architecture: cada modulo tem bindings/, controllers/, views/
- 2. 19 modulos: 5 landing + Login + 17 sidebar (dashboards, silos, secagem, custos, etc)
- 3. Rotas protegidas via AuthMiddleware; publicas sem middleware
- 4. Auth token em FlutterSecureStorage; isAuthenticated controla acesso
- 5. Dio interceptor: 401 = auto-logout + redirect para /login
- 6. GetStorage persiste preferencias (tema, cor primaria) localmente
- 7. Dois endpoints de API: localhost:8000 (Django) + ai.secagemdigital.com (AI/SSE)
- 8. Code generation: freezed + json_serializable + build_runner
- 9. Tema: Material 3, Google Fonts Inter + Outfit, Dark/Light/System
- 10. PDF via pdf + printing packages para geracao de relatorios
- 11. 3D silo visualization via model_viewer_plus (GLB models)
- 12. Localizacao: pt_BR (Portugues Brasil) - unico idioma suportado

Estrutura de diretorios (lib/):